

Sarthak Bichhawa

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Summary

Senior Data Scientist (9+ years) specializing in experimentation, causal measurement, and online learning at cloud scale —adaptive testing programs, multi-touch attribution, and metric design—with impact from 10+ prevented Sev1/Sev2 incidents to ~\$45M/month in recovered revenue.

Experience

Microsoft, Redmond, WA | *Sr. Data & Applied Scientist*

Nov 2020 – Current

Experimentation & Adaptive Testing:

- Built an adaptive pre-release **stress-testing program** for Azure Storage informing per-build ship/hold decisions; helped prevent **10+ Sev1/Sev2 customer incidents** by catching top-severity regressions before rollout.
- Ran **thousands of controlled simulations per build** across 20 workload types, distilling ~500 failure signals into **15 recurring failure modes** and confirming 50+ issues with reproduction steps and quantified impact.
- Prioritized high-yield test scenarios with **online learning (multi-armed bandits)** and tuned scoring thresholds on outcomes, cutting false positives **60%**; led a 2-person effort and authored the design doc + internal white paper.

Causal Measurement & Attribution:

- Recovered and sustained **~\$45M/month in influenced partner revenue** (<1% variance vs Finance, ~4× partner coverage) by rebuilding a 20+ component attribution pipeline with **multi-partner credit** replacing last-writer-wins —lifting partner-attributed influenced revenue **>10% month-over-month**.

Metrics & Customer Health:

- Drove the **Mission-Critical Customers (MCC) health program** and a unified customer-health view across **8+ engineering/support teams**, cutting executive/customer readiness insight from **3-5 days to near real-time**—with ~50% fewer >10-day aging issues and **~16% faster time-to-mitigate**.
- Helped reverse churn/downtrend risk for **10+ critical customers**—each representing multi-millions in monthly Azure Storage revenue—by accelerating early detection and action on customer-impacting issues.

AI-Assisted Analytics:

- Architected an enterprise **LLM agent** that turns natural-language questions into governed live Kusto/ADX queries and PM-ready narratives, cutting customer-health time-to-insight from **days to <2 minutes** across 13 metric domains.

Walmart Labs, Bentonville, AR | *Senior Data Scientist*

Feb 2019 – Nov 2020

- Delivered **~7% category revenue lift** across 50 stores with an end-to-end **planogram optimization** system, cutting planning cycles from weeks to hours.
- Built **product association + substitution analytics** (behavior tree + demand-transfer “universes”) to support assortment and replenishment decisions.

Integral Ad Science, New York, NY | *Data Scientist Intern*

June 2018 – Aug 2018

- Built a PySpark contextual text-classification pipeline processing **~1M URLs/day across 846 categories** (88% accuracy); improved label quality via correlation analysis.

Mu Sigma Business Solutions, Bengaluru, India | *Data Scientist*

Oct 2015 – July 2017

- Estimated **customer lifetime value (CLTV)** with probabilistic/regression models; improved competitive reporting by unifying product IDs via fuzzy matching.

Education

University of Illinois at Chicago (UIC), Chicago, IL | *MS-Business Analytics*

Aug 2017 – Dec 2018

SGSITS, Indore, India | *B.E. in Computer Engineering*

Aug 2011 – July 2015

Skills

- Analytical Techniques:** Causal inference & experimentation (A/B, multi-armed bandits, observational), experiment & metric design, statistical modeling/validation, online learning, clustering & segmentation, anomaly detection, generative & agentic AI
- Analytical Languages & Tools:** Python, SQL, Azure Data Explorer (Kusto), PySpark, R, FastAPI, Power BI, Tableau, Azure DevOps, Git
- Functional Areas:** Experimentation platforms & adaptive testing, causal measurement & multi-touch attribution, marketplace & customer analytics, partner/revenue analytics, production ML